Physical AI SDK - Model Lifecycle & Strategy 2026

Slide 1: Title Slide

Physical AI SDK - Model Lifecycle & Strategy

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Graphic Suggestion: Same futuristic tech cityscape with neural network nodes as original presentation

Slide 2: Model Lifecycle Overview

The Four-Stage Model Lifecycle

```
ENABLEMENT    ->   PROFILING      ->   BENCHMARKING   ->   FINE-TUNING
  Initial          Performance         Validation          Optimization
  Support          Analysis            & Metrics           & Adaptation
```

Key Phases:

1. **Enablement** - Initial model support, integration, basic functionality
2. **Profiling** - Performance analysis, bottleneck identification, hotspot analysis
3. **Benchmarking** - KPI validation, performance metrics, competitive analysis

Slide 3: Progressive Model Support Strategy

From Individual Models to System Orchestration

Phase 1: Individual Model Support

- Focus on single model enablement and optimization
- Establish baseline performance for each model
- Build reusable infrastructure and tooling

Phase 2: Multi-Model Orchestration

Slide 4: Effective Team Collaboration

Progressive Engagement Model

```
Internal Team (PAVS)              →     External Teams (SOW/MCW)
    ↓                                        ↓
Enablement + Initial Profiling          Fine-tuning + Optimization
    ↓                                        ↓
Foundation Building                       Advanced Optimization
```

Strategy:

- **PAVS Team:** Handle enablement and initial profiling
  - Establish working baseline
  - Identify optimization opportunities

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Slide 5: Speed Matters - Parallel Execution

Current Model Status Across Lifecycle Stages

Model Name	Category	Enablement	Profiling	Benchmarking	Fine-tuning
MobileSAM	Vision/Segmentation	✅	✅	✅	🔄
YoloV12	Object Detection	✅	✅	✅	⏸️
Qwen3-VL	VLM	✅	✅	🔄	⏸️
LLaMA3.2-Vision	VLM	✅	✅	⏸️	⏸️
Whisper	ASR	✅	✅	✅	✅

Slide 6: Environment & CI/CD Infrastructure

Isolated Environments for Each Model/System

Environment Strategy:

```
physical-ai-sdk/
├── models/
│   ├── mobilesam/
│   │   ├── Dockerfile
│   │   ├── pyproject.toml (uv)
│   │   └── .venv/
│   ├── yolov12/
│   │   ├── Dockerfile
│   │   ├── requirements.txt (pip)
```

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Slide 7: Container & Virtual Environment Options

Deployment Flexibility

Option 1: Dockerfile (Containerized)

```
FROM rocm/pytorch:latest
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
CMD ["python", "inference.py"]
```

Option 2: uv + pyproject.toml (Modern Python)

```
[project]
name = "mobilesam-inference"
```

Slide 8: Expanding Model Portfolio

Staying Current with Latest Open Source Models

Emerging Neural Alternatives:

- 🔄 **End-to-End Neural Pipelines** replacing traditional CV pipelines
    - Example: Neural SLAM replacing traditional SLAM
    - Example: Neural 3D reconstruction replacing classical methods
- 🆕 **Latest Model Categories to Track:**

Slide 9: Benchmarking & Documentation Strategy

GitHub Pages Integration

Documentation as Code:

```
docs/
├── index.md                        # Landing page
├── models/
│   ├── mobilesam.md               # Model-specific docs
│   ├── yolov12.md
│   └── benchmarks/
│       ├── robotics.md            # Benchmark results
│       └── industrial.md
├── guides/
│   ├── quick-start.md
│   └── deployment.md
└── _config.yml                    # GitHub Pages config
```

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Slide 10: GitHub Pages Example - Mock Preview

Sample Documentation Site Layout

Homepage Preview:

```
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        Physical AI SDK Documentation
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━

┌─────────────────────────────────────────────┐
│  🚀 Quick Start  |  📊 Benchmarks  |  🤖 Models  |
└─────────────────────────────────────────────┘

📚 Model Catalog
├── Vision Models (12)
│   └── MobileSAM, YoloV12, EfficientNet...
├── VLM Models (5)
│   └── Qwen3-VL, LLaMA3.2-Vision...
│
```

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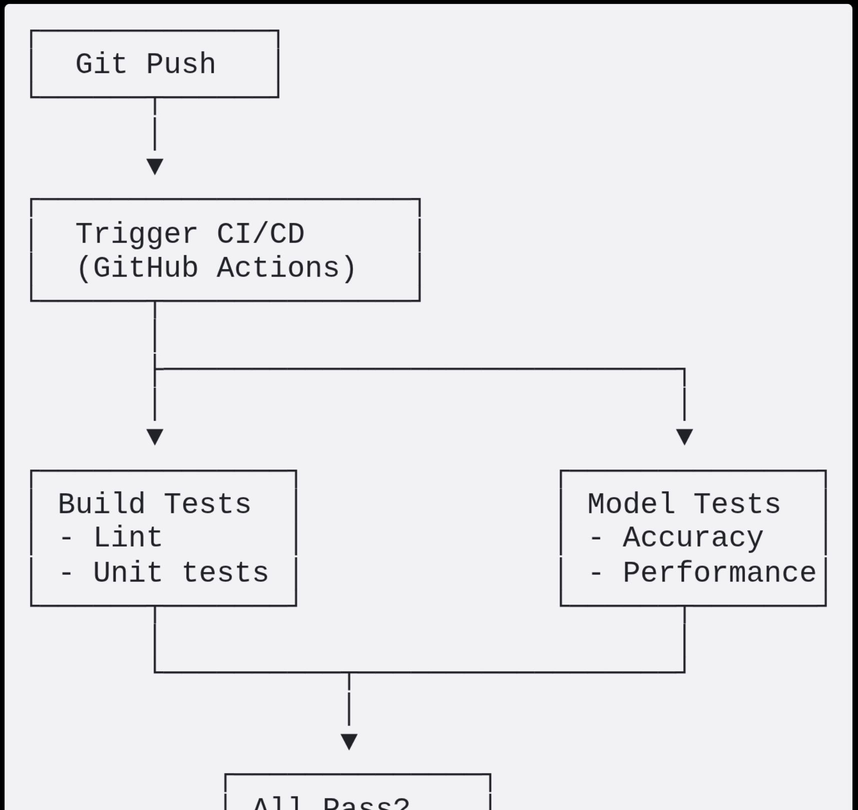
Slide 11: CI/CD Pipeline Detail

Preventing Regressions

Automated Testing Pipeline:



Slide 12: Practical Example - Model Lifecycle

Case Study: Qwen3-VL Integration

Week 1-2: Enablement

- ✅ Model weights download and conversion
- ✅ Basic ONNX-RT integration
- ✅ Simple inference test (single image)
- ✅ Docker environment setup

Week 3-4: Profiling

- 📊 Layer-wise performance analysis (AMD Profiler)

Slide 13: Resource Management Strategy

Balancing Internal & External Resources

Resource Allocation Matrix:

Lifecycle Stage	Internal Team (PAVS)	External Partners	Duration
Enablement	███████████ (Primary)	░░░ (Support)	2-3 weeks
Profiling	████████ (Lead)	████ (Assist)	2-4 weeks
Benchmarking	█████ (Guide)	███████ (Execute)	3-4 weeks
Fine-tuning	███ (Review)	██████████ (Primary)	4-8 weeks

Handoff Criteria:

✓ Model runs successfully on target hardware

✓ Profiling report completed

Slide 14: Technology Watchlist 2026

Emerging Models & Technologies

High Priority:

1. **Qwen3-VL 32B** - Latest multimodal foundation model
2. **RT-X** - Robotics transformer with broader action space
3. **Neural SLAM** - End-to-end learned SLAM systems
4. **Depth Anything V2** - Improved monocular depth estimation

Medium Priority:

5. **SAM 2** - Segment Anything Model v2 with video support

Slide 15: Success Metrics & KPIs

Measuring Lifecycle Effectiveness

Model Lifecycle KPIs:

Metric	Target	Current	Trend
Average Time to Enablement	< 2 weeks	2.5 weeks	⬇️ Improving
Models in Profiling Stage	5-8 concurrent	6	➡️ Stable
Benchmark Pass Rate (1st try)	> 80%	75%	⬆️ Improving
Fine-tuning Speedup (avg)	> 2x	2.3x	⬆️ Good
CI/CD Pipeline Success Rate	> 95%	97%	✅ Excellent

Portfolio Metrics:

- Total Models Supported: **45+**
- Production-Ready Models: **28**

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Slide 16: Roadmap & Next Steps

2026 Execution Plan

Q1 2026 (Current):

- ✅ Establish lifecycle framework
- 🔄 Complete 8 models through full lifecycle
- 🔄 Deploy GitHub Pages documentation

Q2 2026:

- 📋 Expand to 15 production-ready models
- 📋 Implement automated benchmarking CI/CD
- 📋 Onboard 2 external partner teams (MCW/SOW)

Slide 17: Call to Action

Building the Future of Physical AI

Our Commitment:

- 🎯 **Systematic** approach to model support
- 🎯 **Efficient** collaboration with partners
- 🎯 **Rapid** adaptation to emerging technologies
- 🎯 **Quality** focus through CI/CD and benchmarking

Get Involved:

- ✉️ Contact: zhaohui.yan@amd.com
- 🌐 Documentation: [GitHub Pages URL]

Appendix: Technical Architecture

(Include slide 2 from original presentation showing the full SW stack)

Appendix: Model Details by Segment

(Include slides 3-5 from original presentation showing model catalogs)

Graphics & Images Summary

Recommended Image Types for Each Slide:

1. **Title:** Futuristic cityscape with neural networks
2. **Lifecycle:** Circular process diagram with icons
3. **Orchestration:** Before/after architecture diagram
4. **Collaboration:** Partnership/handshake visual
5. **Speed:** Gantt chart / parallel execution visual
6. **Environment:** File tree + CI/CD pipeline
7. **Container:** Technology logos (Docker, Python, UV)
8. **Emerging:** Radar/scanning technology visual
9. **GitHub Pages:** Documentation workflow diagram